

Luke Bhan

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RESEARCH Machine learning, Neural operators, Learning-based control.

CURRENT **University of California San Diego**, La Jolla, California Sept 2022 - Present
EDUCATION *Ph.D. Student, Department of Electrical and Computer Engineering*
 Advisors: Yuanyuan Shi, Miroslav Krstic

PREVIOUS **Vanderbilt University**, Nashville, TN Sept 2020 - May 2022
EDUCATION *Masters of Science, Department of Computer Science*
 Advisor: Guatam Biswas
 Thesis: Deep Reinforcement Learning for Adaptive Control in Robotics

Vanderbilt University, Nashville, TN Sept 2018 - May 2020
 Bachelors of Science, Department of Computer Science and
 Department of Physics and Astronomy
 Advisors: Guatam Biswas and Kálmán Varga

HONORS AND **Best Paper Award Finalist**, Learning for Dynamics and Control Confer-
AWARDS ence 2025 (3/119. For the work [C10] below)

Commitment to Diversity Award, University of California, San Diego,
 Department of Electrical and Computer Engineering (10k funding)

Underwood Memorial Award, Vanderbilt University, Department of
 Physics and Astronomy, (Outstanding graduating senior in physics)

Best Undergraduate Paper Award, Vanderbilt University, Department
 of Physics and Astronomy, (For the work [J1] below)

GRANTS AND **Computational Science Graduate Fellowship**, Department of Energy,
FELLOWSHIPS USA (4 year Ph.D. funding, 400k)

Nvidia Academic Grant, Co-Investigator with Yuanyuan Shi, (8x RTX
 PRO 6000 Server Edition GPUs to support *PDEControlGym: Physical AI*
 Benchmarks for RL Control of PDE Systems)

Ph.D. Fellowship, University of California, San Diego, Department of
 Electrical and Computer Engineering (66k funding)

JOURNAL [J10] Luke Bhan, Miroslav Krstić, Yuanyuan Shi. "Stabilization of non-
PUBLICATIONS linear systems with unknown delays via delay-adaptive neural operator ap-
 proximate predictors." *Automatica, provisionally accepted*,, 2026.

 [J9] Luke Bhan, Peter Quawas, Miroslav Krstic, Yuanyuan Shi. "Sampling-
 Horizon Neural Operator Predictors for Nonlinear Control Under Delayed
 Inputs." *IEEE Control Systems Letters, Volume 10, 2026*.

 [J8] Maxence Lamarque, Luke Bhan, Yuanyuan Shi, Miroslav Krstic. "Adap-

tive Neural-Operator Backstepping Control of a Benchmark Hyperbolic PDE.” *Automatica*, Volume 177, 2025.

[J7] Maxence Lamarque, Luke Bhan, Rafael Vazquez, Miroslav Krstic. ”Gain Scheduling with a Neural Operator for a Transport PDE with Nonlinear Recirculation.” *IEEE Transactions on Automatic Control*, Volume 70. 2025.

[J6] Luke Bhan, Yuanyuan Shi, Miroslav Krstic. ”Adaptive control of reaction–diffusion PDEs via neural operator-approximated gain kernels.” *System & Control Letters*, Volume 195. 2024.

[J5] Miroslav Krstic, Luke Bhan, Yuanyuan Shi. ”Neural operators of backstepping controller and observer gain functions for reaction–diffusion PDEs.” *Automatica*, Volume 164. 2024.

[J4] Luke Bhan, Yuanyuan Shi, Miroslav Krstic. ”Neural operators for bypassing gain and control computations in PDE backstepping.” *IEEE Transactions on Automatic Control*, Volume 69. 2023.

[J3] Luke Bhan, Cody L Covington, Kálmán Varga. ”Laser-Driven Peta-hertz Electron Ratchet Nanobubbles.” *Nano Letters*, Volume 22. 2022.

[J2] Luke Bhan, Cody L Covington, Kálmán Varga. ”Signatures of atomic structure in subfemtosecond laser-driven electron dynamics in nanogaps.” *Physical Review B*, Volume 105. 2022.

[J1] Luke Bhan, Cody L Covington, Jason Rivas, Kálmán Varga. ”Simulation of photo-electron spectrum and electron scattering by dual time propagation.” *The Journal of Chemical Physics*, Volume 154. 2021.

CONFERENCE
PUBLICATIONS

[C12] Filip Bajraktari, Luke Bhan, Miroslav Krstic, Yuanyuan Shi. ”Delay compensation of multi-input distinct delay nonlinear systems via neural operators.” In Proceedings of *American Control Conference (ACC)*, 2026.

[C11] Luke Bhan, Miroslav Krstic, Yuanyuan Shi. ”Delay-adaptive Control of Nonlinear Systems with Approximate Neural Operator Predictors.” In Proceedings of *IEEE Conference on Decision and Control (CDC)*, 2025.

[C10] Luke Bhan*, Peijia Qin*, Miroslav Krstic, Yuanyuan Shi. ”Neural Operators for Predictor Feedback Control of Nonlinear Delay Systems.” In Proceedings of *Learning for Dynamics and Control (L4DC)*, 2025. **Best Paper Finalist.**

[C9] Sharath Matada, Luke Bhan*, Yuanyuan Shi, Nikolay Atanasov. ”Generalizable Motion Planning via Operator Learning.” In Proceedings of *International Conference on Learning Representations (ICLR)*, 2025.

[C8] Luke Bhan*, Yuexin Bian*, Miroslav Krstic, Yuanyuan Shi. ”PDE Control Gym: A Benchmark for Data-Driven Boundary Control of Partial Differential Equations.” In Proceedings of *Learning for Dynamics and Control (L4DC)*, 2024.

[C7] Luke Bhan, Yuanyuan Shi, Iasson Karafyllis, Miroslav Krstic, James B Rawlings. ”Moving-Horizon Estimators for Hyperbolic and Parabolic PDEs in 1-D.” In Proceedings of *American Control Conference (ACC)*, 2024.

[C6] [Luke Bhan](#), Yuanyaun Shi, Miroslav Krstic. "Neural Operators for Hyperbolic PDE Backstepping Feedback Laws." In Proceedings of *IEEE Conference on Decision and Control* (CDC), 2023.

[C5] [Luke Bhan](#), Yuanyaun Shi, Miroslav Krstic. "Neural Operators for Hyperbolic PDE Backstepping Kernels." In Proceedings of *IEEE Conference on Decision and Control* (CDC), 2023.

[C4] [Luke Bhan](#), Yuanyaun Shi, Miroslav Krstic. "Operator learning for nonlinear adaptive control." In Proceedings of *Learning for Dynamics and Control* (L4DC), 2023.

[C3] [Luke Bhan](#), Marcos Quinones-Grueiro, Gautam Biswas. "Concurrent policy blending and system identification for generalized assistive control." In Proceedings of *International Conference on Robotics and Automation* (ICRA), 2022.

[C2] [Luke Bhan](#), Marcos Quinones-Grueiro, Gautam Biswas. "Fault tolerant control combining reinforcement learning and model-based control." In Proceedings of *International Conference on Control and Fault-Tolerant Systems* (SysTol), 2021.

[C1] Adam Stager, [Luke Bhan](#), Andreas Malikopoulos, Liuhui Zhao. "A Scaled Smart City for Experimental Validation of Connected and Automated Vehicles." In Proceedings of *IFAC Symposium on Control in Transportation Systems* (CTS), 2018.

PREPRINTS

Miroslav Krstic, [Luke Bhan](#). "Families of Control-Cost-Parametrized Inverse-Optimal Universal Stabilizers." Submitted to *System & Control Letters*.

Miroslav Kristić, Iasson Karafyllis, [Luke Bhan](#), Carina Veil. "Lotka-Sharpe Neural Operators for Control of Population PDEs." Submitted to *IEEE Transactions on Automatic Control* (TAC).

[Luke Bhan](#), Miroslav Kristić, Yuanyuan Shi. "Predictor-Based Output-Feedback Control of Linear Systems with Time-Varying Input and Measurement Delays via Neural-Approximated Prediction Horizons." Submitted to *System & Control Letters*.

[Luke Bhan](#), Miroslav Krstić, Yuanyuan Shi. "Neural operators for predictor feedback control of nonlinear delay systems." Submitted to *IEEE Transactions on Automatic Control* (TAC).

INTERNSHIPS

Research Scientist Intern, Microsoft June 2026 - Current
On-going. Developing foundation models for the power grid.

Applied Scientist Intern, Amazon June 2025 - Aug 2025
Fine-tuned and designed custom LLMs to provide adjustments for time-series forecasts predicting demands of over 30+ million Amazon products. Demonstrated that LLMs can effectively identify poor distributional forecasts which led to an internal Amazon Machine Learning Conference (AMLC) paper.

Research Intern, Lawrence Berkeley National Lab June 2024 - Aug 2024
Developed randomized linear algebra (RandLA) solvers for distributed non-convex optimization (SQP) enabling real-time, parallel optimization (1x speedup per node) for control of dynamical systems.

	Software Engineering Intern, <i>Mongo DB</i> Created the compression algorithm for MongoDB time-series database.	June 2021 - Aug 2021
	Machine Learning Intern, <i>T-Mobile</i> Created an internal analytics API for visualizing network loads to proactively identify and combat downtime.	June 2021 - Aug 2021
TEACHING	Co-Instructor Physics Informed Machine Learning, University of California San Diego. • Taught 5 Lectures on Foundation Models, Neural ODEs, Neural Operators, and Physics Informed Neural Networks (PINNs). • Created and developed custom homework from scratch exploring augmented neural ODEs, FNOs, and PINNs. Teaching Assistant <i>Numerical Analysis</i> , Vanderbilt University Mathematics Department Teaching Assistant <i>Introduction to Probability and Statistics</i> , Vanderbilt University Mathematics Department Teaching Assistant, <i>Intermediate Software Design</i> Vanderbilt University Computer Science Department	Spring 2026, Spring 2025 Fall 2021 Fall 2021 Fall 2020 - Fall 2021
INVITED TALKS	Seminar Presentations Seminar, University of Washington (Hosted by Prof. Baosen Zhang) Seminar, University of California, San Diego (Hosted by Prof. Sylvia Herbert) Fed-NeMO Seminar, Yale University, (Hosted by Prof. Lu Lu) Seminar, International Institute of Computer Science, UC Berkeley (Michael Mahoney's Group) Seminar, Hong Kong University of Science and Technology (Huan Yu's Group) Internal Control Seminar, University of California, San Diego (Hosted by Yang Zheng) Conference Presentations Paper presentation, American Control Conference (ACC) Paper presentation, IEEE Conference on Decision and Control (CDC) Poster presentation, Learning for Dynamics and Control (L4DC) Paper presentation, International Conference on Robotics and Automation (ICRA) Paper presentation, International Conference on Control and Fault-Tolerant Systems (SyStol)	March. 2026 Sept. 2025 Oct. 2024 June 2024 March 2024 April. 2023 July 2024 Dec. 2024 June 2023 June 2022 Feb. 2022
ACADEMIC SERVICE	Workshop Organizer PDE-AI Workshop: AI for Modeling, Control, and Optimization of Partial Differential Equations, <i>American Control Conference (ACC)</i> (2026). acc-2026-ai-for-pde-workshop.vercel.app Journal Reviewer Automatica (2023-) IEEE Transactions on Automatic Control (2023-) Systems & Control Letters (2024-) International Journal of Robust and Nonlinear Control (2024-) IEEE Control Systems Letters (2025-) IEEE Transactions on Control of Networked Systems (2026-)	

Conference Reviewer

IEEE Conference on Decision and Control (CDC), (2023-)

American Control Conference (ACC), (2024-)

Learning for Dynamics and Control (L4DC), (2022-)

Association for the Advancement of Artificial Intelligence Conference (AAAI),
(2023-)

International Conference on Learning Representations (ICLR), (2023-)